Polar Curves

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1.Polar Coordinates Practice

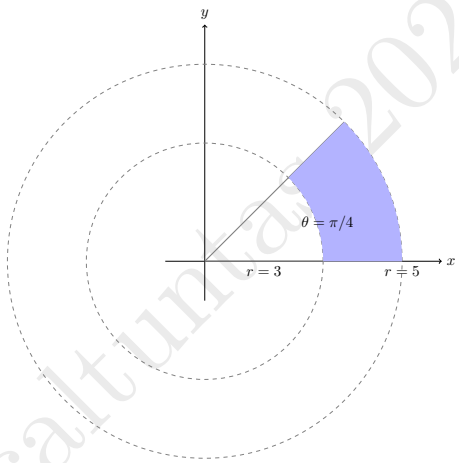
Concept Overview.

Each point in the plane can be represented by a radius r and an angle θ. Regions defined by inequalities in polar coordinates can be visualized using sectors and arcs.

Problem 1. *Graph the following polar regions:*

1. *All points satisfying* $3 \leq r \leq 5$ *and* $0 \leq \theta \leq \pi/4$.
2. *All points satisfying* $r \leq 0$ *and* $\theta = \pi/3$.

Solution. ***1.*** *The first region is a sector of an annulus with inner radius 3, outer radius 5, spanning angles from* 0 *to* $\pi/4$.



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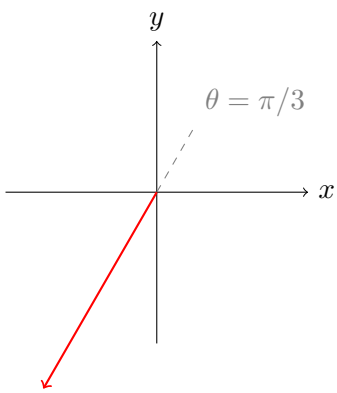
2. *The second region represents points in the direction opposite* $\theta = \pi/3$ *because* r *is negative.*



Problem 2. *Rewrite the following Cartesian equations in equivalent polar form:*

1. $xy = 3$
2. $(x+1)^2 + (y-4)^2 = 9$
3. $x = 5$
4. $x^2 + xy + y^2 = 2$

Solution. *Recall that in polar coordinates $x = r\cos\theta$ and $y = r\sin\theta$. Then:*

1. *For $xy = 3$:*

$$(r\cos\theta)(r\sin\theta) = 3 \implies r^2\cos\theta\sin\theta = 3 \implies r = \sqrt{\frac{3}{\cos\theta\sin\theta}}$$

2. *For $(x+1)^2 + (y-4)^2 = 9$:*

$$\begin{aligned}(x+1)^2 + (y-4)^2 &= 9\\ (r\cos\theta+1)^2 + (r\sin\theta-4)^2 &= 9\\ r^2(\cos^2\theta+\sin^2\theta) + 2r(\cos\theta-4\sin\theta) + (1+16) &= 9\\ r^2 + 2r(\cos\theta-4\sin\theta) + 17 &= 9\\ r^2 &= -2r(\cos\theta-4\sin\theta) - 8\end{aligned}$$

3. *For the vertical line $x = 5$:*

$$r\cos\theta = 5 \implies r = \frac{5}{\cos\theta}$$

4. *For $x^2 + xy + y^2 = 2$:*

$$(r\cos\theta)^2 + (r\cos\theta)(r\sin\theta) + (r\sin\theta)^2 = 2 \implies r^2(\cos^2\theta + \cos\theta\sin\theta + \sin^2\theta) = 2$$

$$r = \sqrt{\frac{2}{1+\cos\theta\sin\theta}}$$

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Problem 3. *Replace the following polar equations by equivalent Cartesian equations:*

1. $r^2 = -6r\cos\theta$
2. $r = \theta$
3. $r = (\csc\theta)e^{r\cos\theta}$
4. $r = 3\cot\theta\csc\theta$
5. $r = 7\sin\theta - 4\cos\theta$

Solution. 1. *For $r^2 = -6r\cos\theta$:*

$$\begin{aligned} r^2 &= -6r\cos\theta \\ r^2 + 6r\cos\theta &= 0 \\ r(r + 6\cos\theta) &= 0 \end{aligned}$$

So either $r = 0$ (the origin) or

$$r + 6\cos\theta = 0 \implies r = -6\cos\theta.$$

Using $x = r\cos\theta$ and $y = r\sin\theta$:

$$r^2 = x^2 + y^2, \quad r\cos\theta = x$$

$$\begin{aligned} r = -6\cos\theta &\implies r = -6\frac{x}{r} \implies r^2 = -6x \\ x^2 + y^2 = -6x &\implies x^2 + y^2 + 6x = 0. \end{aligned}$$

2. *For $r = \theta$: This equation cannot be expressed in a simple Cartesian form because $\theta = \arctan(y/x)$.*

$$r = \sqrt{x^2 + y^2}, \quad \theta = \arctan\frac{y}{x} \implies \sqrt{x^2 + y^2} = \arctan\frac{y}{x}.$$

3. *For $r = (\csc\theta)e^{r\cos\theta}$:*

$$r = \frac{1}{\sin\theta}e^{r\cos\theta} \implies r\sin\theta = e^{r\cos\theta}.$$

Since $r\sin\theta = y$ and $r\cos\theta = x$, we get the Cartesian form:

$$y = e^x.$$

4. *For $r = 3\cot\theta\csc\theta$:*

$$\begin{aligned} \cot\theta = \frac{\cos\theta}{\sin\theta}, &\quad \csc\theta = \frac{1}{\sin\theta} \\ r = 3\frac{\cos\theta}{\sin\theta}\cdot\frac{1}{\sin\theta} &= \frac{3\cos\theta}{\sin^2\theta} \\ r\sin^2\theta = 3\cos\theta &\implies r(\sin^2\theta) = 3\cos\theta \end{aligned}$$

$$r\sin^2\theta = r\sin\theta\cdot\sin\theta = y\cdot\sin\theta \implies r\sin^2\theta = y\frac{y}{r} = \frac{y^2}{r}$$

$$\frac{y^2}{r} = 3\frac{x}{r} \implies y^2 = 3x$$

5. *For $r = 7\sin\theta - 4\cos\theta$:*

$$\begin{aligned} r &= 7\sin\theta - 4\cos\theta \\ r &= 7\frac{y}{r} - 4\frac{x}{r} \implies r^2 = 7y - 4x \\ x^2 + y^2 &= -4x + 7y \end{aligned}$$

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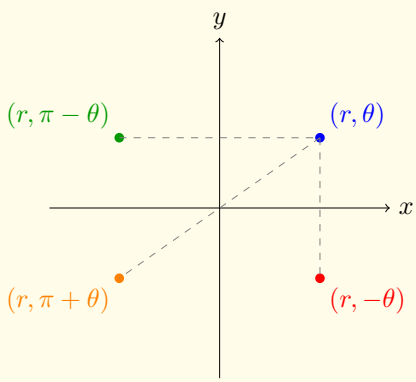
Problem 4. *Graph the following polar curves:*

1. $r = 1 + \cos\theta$
2. $r = 2\sin\theta$
3. $r^2 = \cos 2\theta$

Hint

Symmetry Principles:
If (r, θ) is on the curve, then:

- **x-axis symmetry:** If $(r, -\theta)$ or $(-r, \pi - \theta)$ is on the curve.
- **y-axis symmetry:** If $(r, \pi - \theta)$ or $(-r, -\theta)$ is on the curve.
- **Origin symmetry:** If $(-r, \theta)$ or $(r, \pi + \theta)$ is on the curve.



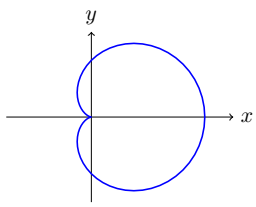
Solution. ***1.*** $r = 1 + \cos\theta$ ***(Cardioid)***

Step 1: Symmetry. *Replacing θ with $-\theta$ gives $r = 1 + \cos(-\theta) = 1 + \cos\theta$. Thus, the curve is symmetric about the* ***x-axis****.*

Step 2: Table of Points.

θ	0	$\pi/3$	$\pi/2$	$2\pi/3$	π
r	2	1.5	1	0.5	0

Step 3: Plot.



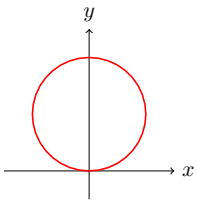
2. $r = 2\sin\theta$ ***(Circle)***

Step 1: Symmetry. *Replacing θ with $\pi - \theta$ gives $r = 2\sin(\pi - \theta) = 2\sin\theta$. Thus, the curve is symmetric about the* ***y-axis****.*

Step 2: Table of Points.

θ	0	$\pi/6$	$\pi/4$	$\pi/3$	$\pi/2$
r	0	1	$\sqrt{2} \approx 1.41$	$\sqrt{3} \approx 1.73$	2

Step 3: Plot.



Problem 5. *Find the equations for the horizontal and vertical tangent lines to the following curves:*

1. $r = 1 + \sin\theta, \quad 0 \le \theta \le 2\pi$
2. $r = 1 - \cos\theta, \quad 0 \le \theta \le 2\pi$

Solution. *The slope of the tangent line is given by* $\frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta}$, *where* $y = r\sin\theta$ *and* $x = r\cos\theta$*:*

$$\frac{dy}{dx} = \frac{f'(\theta)\sin\theta + f(\theta)\cos\theta}{f'(\theta)\cos\theta - f(\theta)\sin\theta}$$

1. Curve: $r = 1 + \sin\theta$
Step 1: Parametric derivatives

- $y = (1 + \sin\theta)\sin\theta = \sin\theta + \sin^2\theta$
- $\frac{dy}{d\theta} = \cos\theta + 2\sin\theta\cos\theta = \cos\theta(1 + 2\sin\theta)$
- $x = (1 + \sin\theta)\cos\theta = \cos\theta + \sin\theta\cos\theta$
- $\frac{dx}{d\theta} = -\sin\theta + \cos 2\theta = -\sin\theta + (1 - 2\sin^2\theta) = -(2\sin\theta - 1)(\sin\theta + 1)$

Step 2: Horizontal Tangents ($\frac{dy}{d\theta} = 0$)

- $\cos\theta(1 + 2\sin\theta) = 0 \implies \theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$
- *At* $\theta = \frac{\pi}{2} : y = (1 + 1)(1) = 2$
- *At* $\theta = \frac{7\pi}{6}, \frac{11\pi}{6} : y = (1 - 1/2)(-1/2) = -1/4$
- *(At* $\theta = 3\pi/2$*, both derivatives are zero; there is a cusp at the pole).*

Step 3: Vertical Tangents ($\frac{dx}{d\theta} = 0$)

- $(2\sin\theta - 1)(\sin\theta + 1) = 0 \implies \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$
- *At* $\theta = \frac{\pi}{6} : x = (1 + 1/2)(\sqrt{3}/2) = \frac{3\sqrt{3}}{4}$
- *At* $\theta = \frac{5\pi}{6} : x = (1 + 1/2)(-\sqrt{3}/2) = -\frac{3\sqrt{3}}{4}$
- *At* $\theta = \frac{3\pi}{2} : x = 0$ *(Tangent at the pole)*

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2. Curve: $r = 1 - \cos\theta$
Step 1: Parametric derivatives

- $y = (1 - \cos\theta)\sin\theta \implies \frac{dy}{d\theta} = \cos\theta - \cos 2\theta = -(2\cos\theta + 1)(\cos\theta - 1)$
- $x = (1 - \cos\theta)\cos\theta \implies \frac{dx}{d\theta} = -\sin\theta + \sin 2\theta = \sin\theta(2\cos\theta - 1)$

Step 2: Horizontal Tangents ($\frac{dy}{d\theta} = 0$)

- $(2\cos\theta + 1)(\cos\theta - 1) = 0 \implies \theta = \frac{2\pi}{3}, \frac{4\pi}{3}, 0$
- *At* $\theta = \frac{2\pi}{3} : y = (1 - (-1/2))(\sqrt{3}/2) = \frac{3\sqrt{3}}{4}$
- *At* $\theta = \frac{4\pi}{3} : y = (1 - (-1/2))(-\sqrt{3}/2) = -\frac{3\sqrt{3}}{4}$
- *At* $\theta = 0$, *we have* $dx/d\theta = 0$ *and* $dy/d\theta = 0$. *By calculating* $\lim_{\theta\to 0}\frac{dy}{dx} = 0$, *we find a horizontal tangent line* $y = 0$ *at the cusp.*

Step 3: Vertical Tangents ($\frac{dx}{d\theta} = 0$)

- $\sin\theta(2\cos\theta - 1) = 0 \implies \theta = \pi, \frac{\pi}{3}, \frac{5\pi}{3}, 0$
- *At* $\theta = \pi : x = (1 - (-1))(-1) = -2$
- *At* $\theta = \frac{\pi}{3}, \frac{5\pi}{3} : x = (1 - 1/2)(1/2) = 1/4$
- *At* $\theta = 0 : x = 0$ *(Tangent at the pole)*

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Problem 6. *Find the areas of the following regions:*

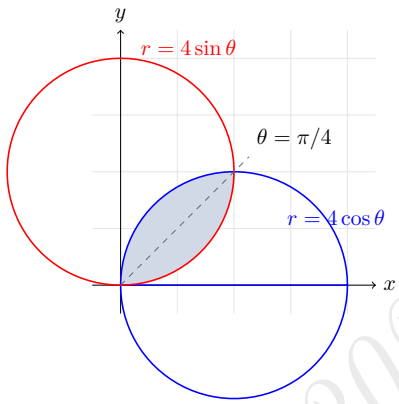
1. *Shared by the circles* $r = 4\cos\theta$ *and* $r = 4\sin\theta$.
2. *Shared by the circle* $r = 4$ *and the cardioid* $r = 4(1 - \cos\theta)$.
3. *Inside the circle* $r = -6\cos\theta$ *and outside the circle* $r = 3$.

Solution. *The area is given by* $A = \int_a^b \frac{1}{2}[r(\theta)]^2 d\theta$.

1. Shared by $r = 4\cos\theta$ ***and*** $r = 4\sin\theta$

Step 1: Intersection. $4\cos\theta = 4\sin\theta \implies \tan\theta = 1 \implies \theta = \pi/4$.

Step 2: Plot and Setup. *Due to symmetry about* $\theta = \pi/4$*:*



$$A = 2\int_0^{\pi/4} \frac{1}{2}(4\sin\theta)^2 d\theta = \int_0^{\pi/4} 16\sin^2\theta d\theta$$

Using the identity $\sin^2\theta = \frac{1-\cos 2\theta}{2}$*:*

$$A = 16\int_0^{\pi/4} \frac{1-\cos 2\theta}{2} d\theta = 8\int_0^{\pi/4} (1-\cos 2\theta) d\theta = 8\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi/4} = 2\pi - 4$$

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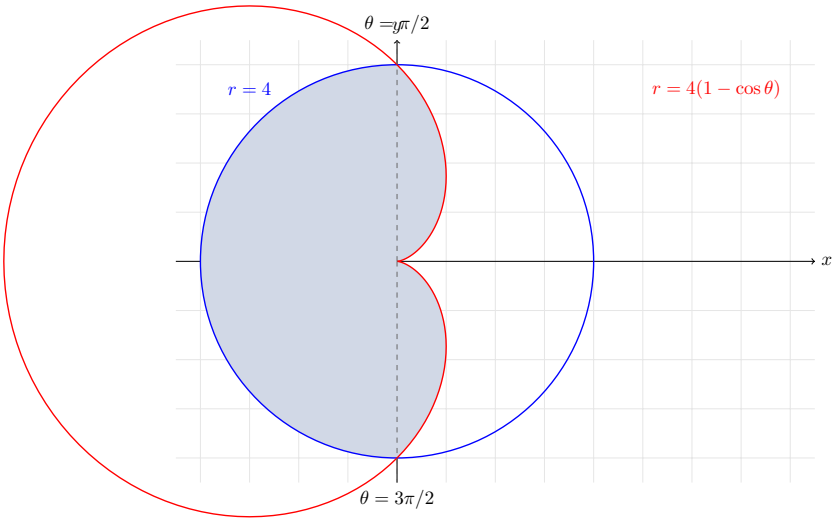
2. Shared by $r = 4$ ***and*** $r = 4(1 - \cos\theta)$

Step 1: Intersection. $4 = 4(1 - \cos\theta) \implies \cos\theta = 0 \implies \theta = \pm\pi/2$.

Step 2: Plot and Setup.



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The area is the sum of the cardioid side and the circle side:

$$A = \int_{-\pi/2}^{\pi/2} \frac{1}{2}[4(1-\cos\theta)]^2 d\theta + \frac{1}{2}(\textit{Area of circle for } \pi/2 \le \theta \le 3\pi/2)$$

Expand $(1-\cos\theta)^2 = 1 - 2\cos\theta + \cos^2\theta$*:*

$$A = 8\int_{-\pi/2}^{\pi/2} (1 - 2\cos\theta + \cos^2\theta) d\theta + 8\pi$$

Use $\cos^2\theta = \frac{1+\cos 2\theta}{2}$ *and combine terms* $(1 + 1/2 = 3/2)$*:*

$$A = 8\int_{-\pi/2}^{\pi/2} \left(\frac{3}{2} - 2\cos\theta + \frac{\cos 2\theta}{2}\right) d\theta + 8\pi$$

Integrating each term:

$$A = 8\left[\frac{3\theta}{2} - 2\sin\theta + \frac{\sin 2\theta}{4}\right]_{-\pi/2}^{\pi/2} + 8\pi = 20\pi - 32$$

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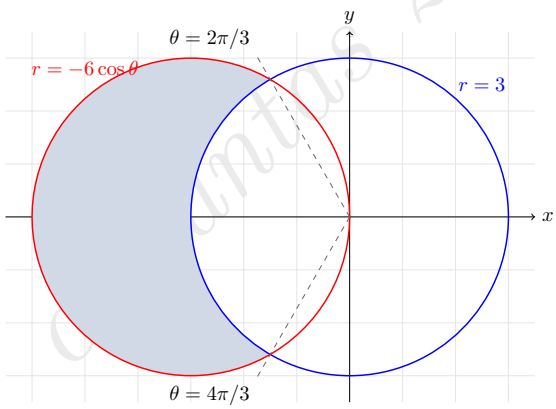
3. Inside $r = -6\cos\theta$ ***and outside*** $r = 3$
Step 1: Intersection. $-6\cos\theta = 3 \implies \cos\theta = -1/2 \implies \theta = 2\pi/3, 4\pi/3.$
Step 2: Plot and Setup.



$$A = \int_{2\pi/3}^{4\pi/3} \frac{1}{2}[(-6\cos\theta)^2 - (3)^2] d\theta = \int_{2\pi/3}^{4\pi/3} (18\cos^2\theta - 4.5) d\theta$$

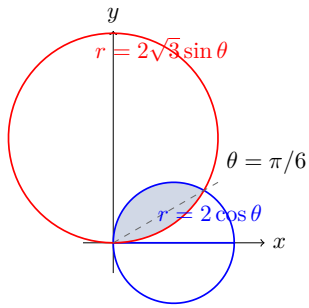
Using $\cos^2\theta = \frac{1+\cos 2\theta}{2}$ *and simplifying* $(18/2 = 9$ *and* $9 - 4.5 = 4.5)$*:*

$$A = \int_{2\pi/3}^{4\pi/3} (4.5 + 9\cos 2\theta) d\theta = [4.5\theta + 4.5\sin 2\theta]_{2\pi/3}^{4\pi/3} = 3\pi + \frac{9\sqrt{3}}{2}$$

Problem 7. *Find the areas of the following polar regions:*

1. *Shared by the circles* $r = 2\cos\theta$ *and* $r = 2\sqrt{3}\sin\theta$*.*
2. *Shared by the circle* $r = 3$ *and the cardioid* $r = 3(1 + \cos\theta)$*.*
3. *Inside the circle* $r = 4$ *and above the line* $r = 2\csc\theta$*.*

Solution. ***1. Shared by*** $r = 2\cos\theta$ ***and*** $r = 2\sqrt{3}\sin\theta$
Step 1: Intersection and Setup *Intersection at* $\theta = \pi/6$*. Area is split:*



$$A = \int_0^{\pi/6} \frac{1}{2}(2\sqrt{3}\sin\theta)^2 d\theta + \int_{\pi/6}^{\pi/2} \frac{1}{2}(2\cos\theta)^2 d\theta$$

$$A = \int_0^{\pi/6} 3(1-\cos 2\theta)d\theta + \int_{\pi/6}^{\pi/2} (1+\cos 2\theta)d\theta$$

$$A = 3\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi/6} + \left[\theta + \frac{\sin 2\theta}{2}\right]_{\pi/6}^{\pi/2} = \frac{5\pi}{6} - \sqrt{3}$$

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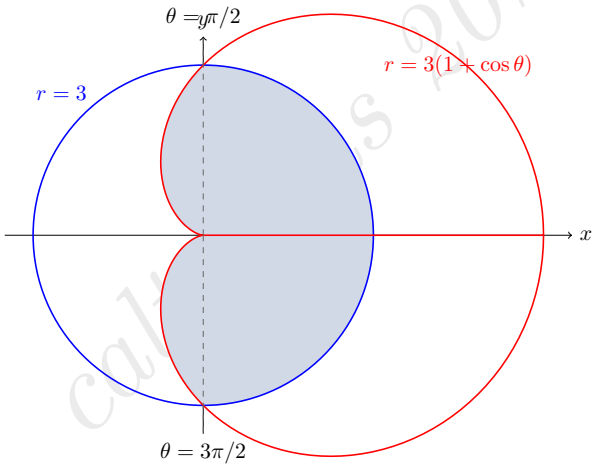
2. Shared by $r = 3$ ***and*** $r = 3(1+\cos\theta)$
Step 1: Intersection and Expansion *Intersection at* $\theta = \pi/2$. *Total area:*



$A = \frac{9\pi}{2} + 9\int_{\pi/2}^{\pi}(1 + 2\cos\theta + \cos^2\theta)d\theta$. *Substitute* $\cos^2\theta = \frac{1+\cos 2\theta}{2}$:

$$A = \frac{9\pi}{2} + 9\int_{\pi/2}^{\pi}\left(\frac{3}{2} + 2\cos\theta + \frac{\cos 2\theta}{2}\right)d\theta = 9\pi - 18$$

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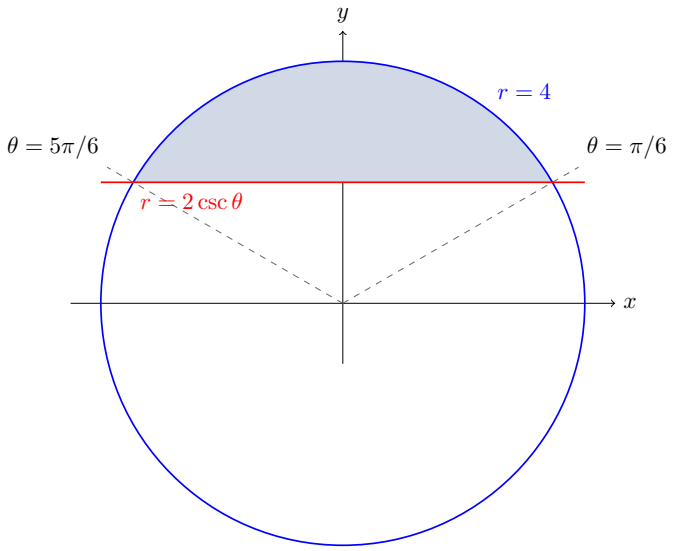
3. Inside $r = 4$ ***and above*** $r = 2\csc\theta$
Step 1: Conversion and Intersection $r\sin\theta = 2 \implies y = 2$. *Intersection at* $\theta = \pi/6, 5\pi/6$.



$$A = \int_{\pi/6}^{5\pi/6} \frac{1}{2}(16 - 4\csc^2\theta)d\theta = [8\theta + 2\cot\theta]_{\pi/6}^{5\pi/6} = \frac{16\pi}{3} - 4\sqrt{3}$$

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Problem 8. *Find the lengths of the following polar curves:*

1. *The spiral* $r = \theta$ *for* $0 \leq \theta \leq 3$.
2. *The circle* $r = a\cos\theta$ *for* $0 \leq \theta \leq \pi$ $(a > 0)$.

Solution. *The arc length* L *of a curve* $r = f(\theta)$ *from* $\theta = \alpha$ *to* $\theta = \beta$ *is:*

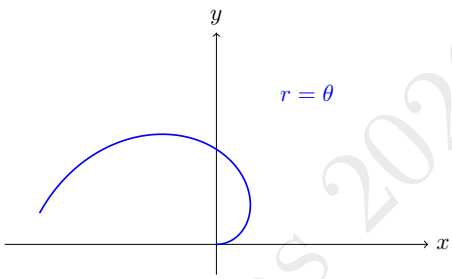
$$L = \int_\alpha^\beta \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2}\, d\theta$$

1. The spiral $r = \theta, \quad 0 \leq \theta \leq 3$

$$r = \theta \implies \frac{dr}{d\theta} = 1$$

$$L = \int_0^3 \sqrt{\theta^2 + 1}\, d\theta$$

(Not needed but here is the plot:)



and the rest is ***HW****.*

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2. The circle $r = a\cos\theta, \quad 0 \leq \theta \leq \pi$

$$r = a\cos\theta \implies \frac{dr}{d\theta} = -a\sin\theta$$

$$L = \int_0^\pi \sqrt{a^2\cos^2\theta + a^2\sin^2\theta}\, d\theta = \int_0^\pi a\, d\theta$$

So,

$$L = [a\theta]_0^\pi = a\pi$$

. Note that for $0 \leq \theta \leq \pi$*, the curve traces the entire circle exactly once.*

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